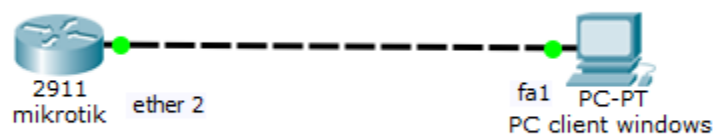


Lab 4

Command Line Interface [terminal] pada mikrotik

Sebelum melakukan konfigurasi nya pastikan mikrotik dan pc/laptop kalian sudah terhubung seperti topologi di bawah ini



Selain menggunakan GUI mikrotik juga di remote melalui via CLI (text mode) mode cli biasanya digunakan oleh expert karena harus sudah mengingat semua command di dalam mikrotik, disini saya akan remote mikrotik menggunakan winbox dan akan mengenalkan kepada kalian command dasar mikrotik.

- Mengganti identitas mikrotik

```
Terminal <1>
MMM MMM MMM III KKK KKK RRRRRR OOOOOO TTT III KKK KK
MMM MM MMM III KKKKK RRR RRR OOO OOO TTT III KKKKK
MMM   MMM III KKK KKK RRRRRR OOO OOO TTT III KKK KKK
MMM   MMM III KKK KKK RRR RRR OOOOOO TTT III KKK KK

MikroTik RouterOS 6.49.13 (c) 1999-2024      http://www.mikrotik.com/

Do you want to see the software license? [Y/n]: n
[?]          Gives the list of available commands
command [?]  Gives help on the command and list of arguments

[Tab]        Completes the command/word. If the input is ambiguous,
              a second [Tab] gives possible options

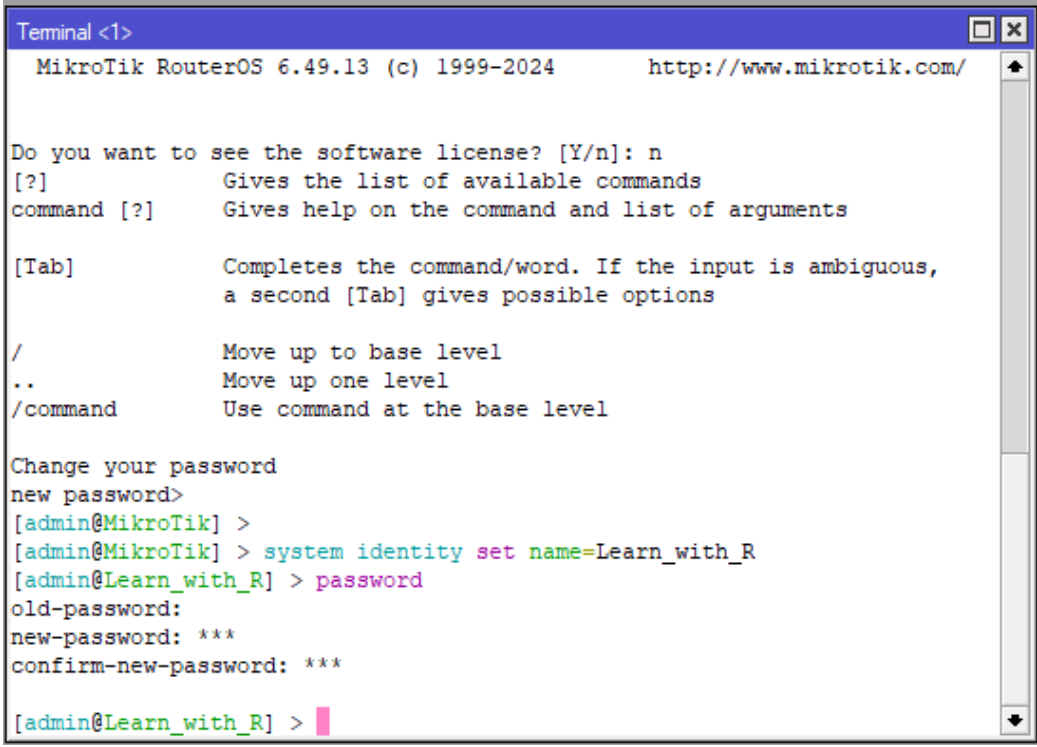
/            Move up to base level
..           Move up one level
/command     Use command at the base level

Change your password
new password>
[admin@MikroTik] >
[admin@MikroTik] > system identity set name=Learn_with_R
[admin@Learn_with_R] >
```

System identity set name=(masukan sesuai keinginan kalian

- Cara ganti Password

Dengan perintah **Password**



```
Terminal <1>
MikroTik RouterOS 6.49.13 (c) 1999-2024      http://www.mikrotik.com/

Do you want to see the software license? [Y/n]: n
[?]          Gives the list of available commands
command [?]  Gives help on the command and list of arguments

[Tab]        Completes the command/word. If the input is ambiguous,
              a second [Tab] gives possible options

/            Move up to base level
..           Move up one level
/command     Use command at the base level

Change your password
new password>
[admin@MikroTik] >
[admin@MikroTik] > system identity set name=Learn_with_R
[admin@Learn_with_R] > password
old-password:
new-password: ***
confirm-new-password: ***

[admin@Learn_with_R] >
```

- Cara membersihkan bekas command di mikrotik

```
Terminal <1>

K
MMM MM MMM III KKKKK RRR RRR OOO OOO TTT III KKKKK
MMM   MMM III KKK KKK RRRRRR OOO OOO TTT III KKK KKK
MMM   MMM III KKK KKK RRR RRR OOOOOO TTT III KKK KK
K

MikroTik RouterOS 6.49.13 (c) 1999-2024      http://www.mikrotik.com/

[?]          Gives the list of available commands
command [?]  Gives help on the command and list of arguments

[Tab]        Completes the command/word. If the input is ambiguous,
              a second [Tab] gives possible options

/            Move up to base level
..           Move up one level
/command     Use command at the base level

[admin@Learn_with_R] > 
```

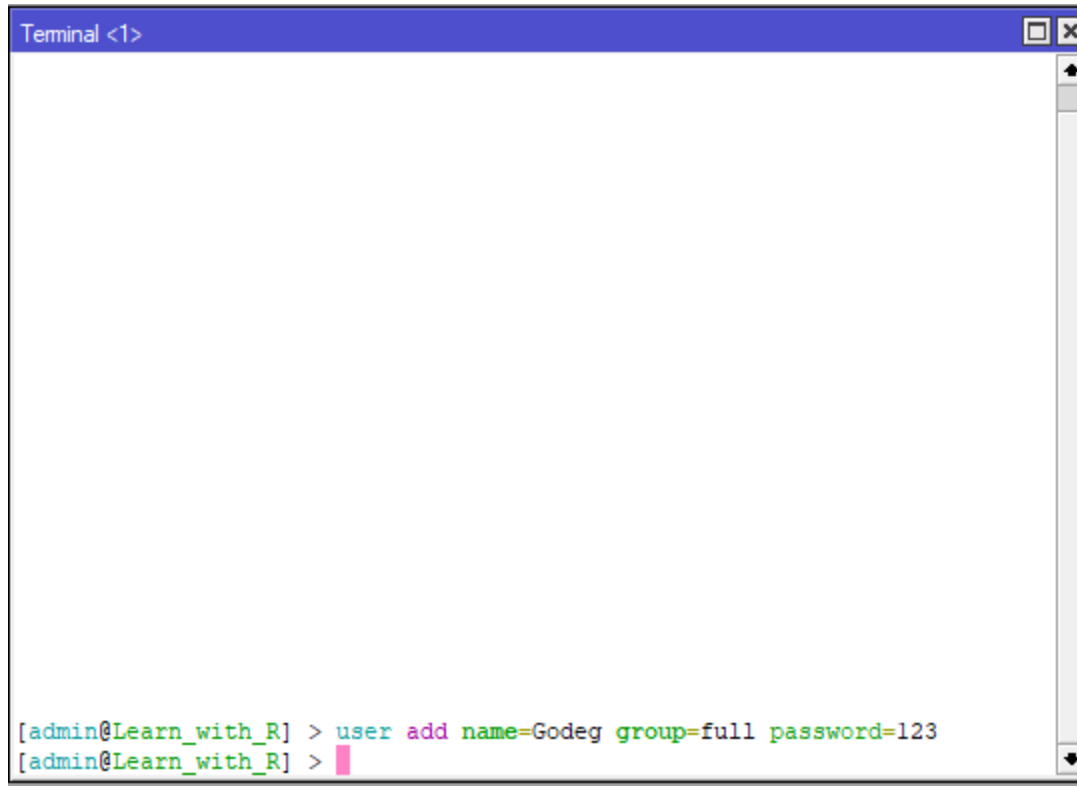
Agar layar nya kembali bersih kalian bisa click **F5** di keyboard

```
Terminal <1>

[admin@Learn_with_R] > 
```

- Membuat user baru dan password dengan hak akses full

Dengan perintah **user add name=(sesuai keinginan kalian) group=full password=123**

A screenshot of a terminal window titled "Terminal <1>". The window has a blue title bar and a white body. At the bottom, there are two lines of text: the first line shows the command `[admin@Learn_with_R] > user add name=Godeg group=full password=123` and the second line shows the prompt `[admin@Learn_with_R] >` followed by a pink cursor. The terminal window has standard window controls (minimize, maximize, close) in the top right corner and a scrollbar on the right side.

```
Terminal <1>

[admin@Learn_with_R] > user add name=Godeg group=full password=123
[admin@Learn_with_R] >
```

- Melihat user
Bisa dengan perintah **User Print**

```
[admin@Learn_with_R] > user print
Flags: E - expired, X - disabled
#   NAME          GROUP      ADDRESS          LAST-LOGGED-IN
0   ;;: system default user
    admin         full      jan/02/1970 00:35:30
1   Godeg         full
[admin@Learn with R] > █
```

- Melihat lisensi

Bisa dengan perintah **system license Print**

```
Terminal <1>
[admin@Learn_with_R] > system license print
software-id: B3L3-TIVD
nlevel: 4
features:
[admin@Learn_with_R] > █
```

- Melihat nama interface dan mengganti nama interface

Perintah mengganti : **interface set name** =(sesuai keinginan kalian)

Perintah melihat nama interface : **interface print**

```
Terminal <1>
expected end of command (line 1 column 37)
[admin@Learn_with_R] > interface set name=ether1-WAN ether1
[admin@Learn_with_R] > interface set name=ether2-WAN ether2
[admin@Learn_with_R] >
[admin@Learn_with_R] > interface print
Flags: D - dynamic, X - disabled, R - running, S - slave
#   NAME          TYPE      ACTUAL-MTU L2MTU
0   R ether1-WAN    ether     1500      1598
1   R ether2-WAN    ether     1500      1598
2   ether3         ether     1500      1598
3   ether4         ether     1500      1598
4   ether5         ether     1500      1598
5   X wlan1        wlan      1500      1600
[admin@Learn_with_R] > 
```

- Memberikan ip address dan melihat ip address

Perintah menambahkan ip : **ip address add address=10.10.10.1/24 interface=ether2**

perintah mengganti ip : **ip address add address=10.10.10.5/24 interface=ether2**

perintah melihat ip address: **ip address print**

```
Terminal <1>

[admin@Learn_with_R] > ip address add address=10.10.10.1/24 interface=ether2-
WAN
[admin@Learn_with_R] > ip address print
Flags: X - disabled, I - invalid, D - dynamic
#   ADDRESS      NETWORK    INTERFACE
0   10.10.10.1/24  10.10.10.0 ether2-WAN
[admin@Learn_with_R] > 
```

```
Terminal <1>

[admin@Learn_with_R] > ip address set address=10.10.10.5/24 interface=ether2-WAN
numbers: 0
[admin@Learn_with_R] > ip address print
Flags: X - disabled, I - invalid, D - dynamic
#   ADDRESS          NETWORK    INTERFACE
0   10.10.10.5/24     10.10.10.0 ether2-WAN
[admin@Learn_with_R] >
```

- Menambahkan nat dan melihat konfigurasi nat

Perintah menambahkan nat: **ip firewall nat add chain=srcnat out-interface=ether1 action=masquerade**

Perintah lihat konfigurasi nat : **ip firewall nat print**

```
Terminal <1>

[admin@Learn_with_R] > ip firewall nat add chain=srcnat out-interface=ether1-WAN action=masquerade
[admin@Learn_with_R] > ip firewall nat print
Flags: X - disabled, I - invalid, D - dynamic
0   chain=srcnat action=masquerade out-interface=ether1-WAN
[admin@Learn_with_R] >
```

- Menambahkan dan melihat gateway

Perintah menambahkan gateway : **ip route add gateway=(gateway ip yang kalian pake)**

```
Terminal <1>

[admin@Learn_with_R] > ip route add gateway=10.10.10.1
[admin@Learn_with_R] > ip route print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
#      DST-ADDRESS      PREF-SRC      GATEWAY      DISTANCE
0 A S  0.0.0.0/0          10.10.10.1    1
1 ADC  10.10.10.0/24      10.10.10.5    ether2-WAN    0
[admin@Learn_with_R] >
```

- Menambahkan dns

Bisa dengan perintah : **ip dns set servers=8.8.8.8 allow-remote-request=yes**

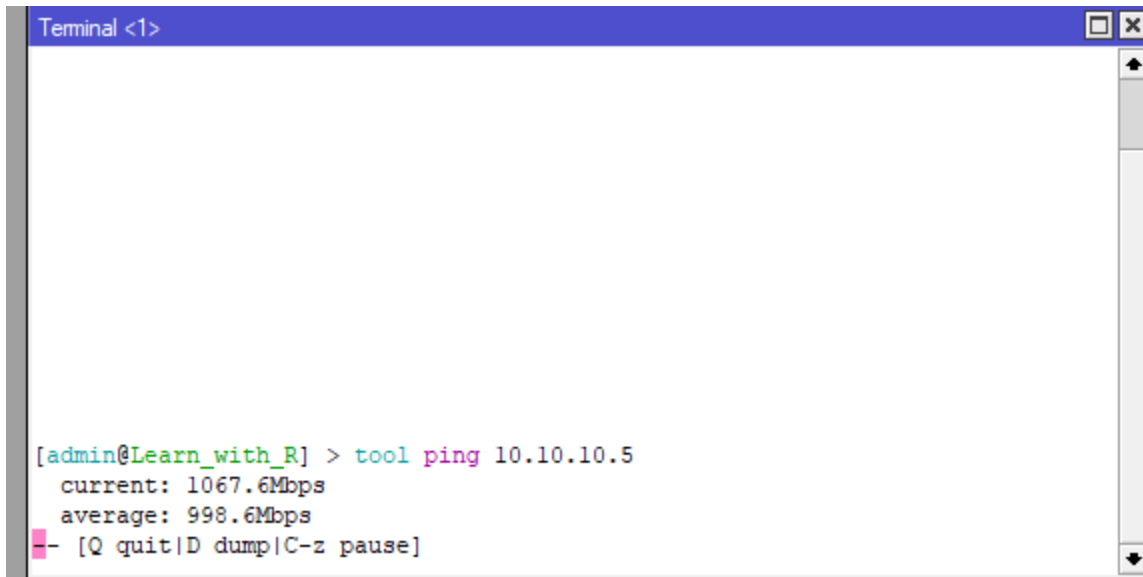
Perintah melihat dns: **ip dns print**

```
Terminal <1>

[admin@Learn_with_R] > ip dns set servers=8.8.8.8 allow-remote-requests=yes
[admin@Learn_with_R] > ip dns print
      servers: 8.8.8.8
      dynamic-servers:
      use-doh-server:
      verify-doh-cert: no
      allow-remote-requests: yes
      max-udp-packet-size: 4096
      query-server-timeout: 2s
      query-total-timeout: 10s
      max-concurrent-queries: 100
      max-concurrent-tcp-sessions: 20
      cache-size: 2048KiB
      cache-max-ttl: 1w
      cache-used: 25KiB
-- [Q quit|D dump]
```


- Melakukan ping:

Perintah nya : tool Ping (ip yang ingin di hubungi)

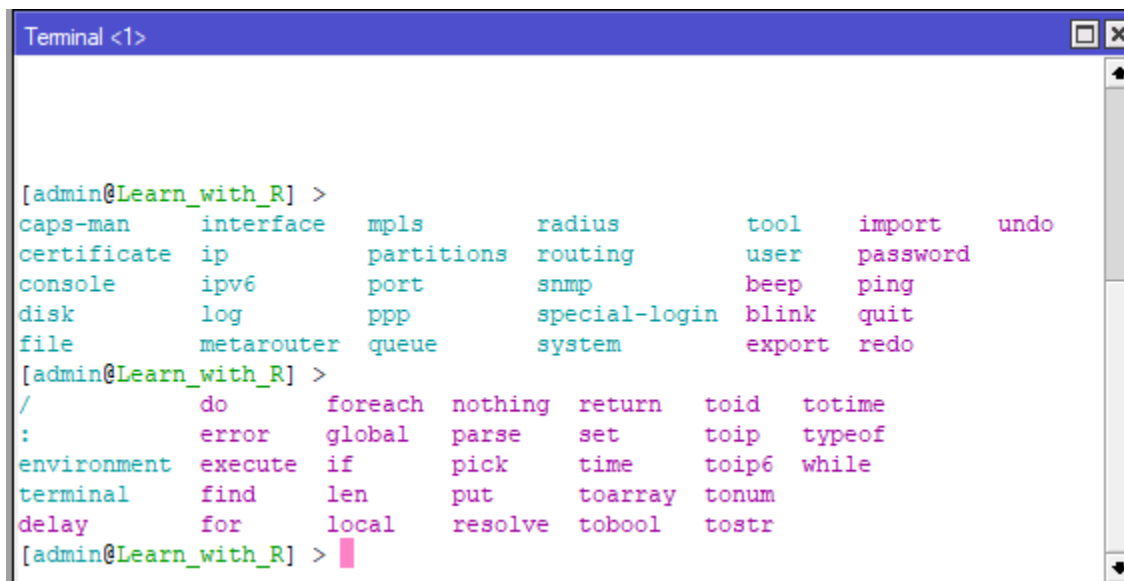


```
Terminal <1>

[admin@Learn_with_R] > tool ping 10.10.10.5
  current: 1067.6Mbps
  average: 998.6Mbps
- [Q quit/D dump/C-z pause]
```

- Perintah untuk melihat perintah selanjut nya:

Tekan Tab di keyboard 2 kali saat bingung melakukan perintah apa

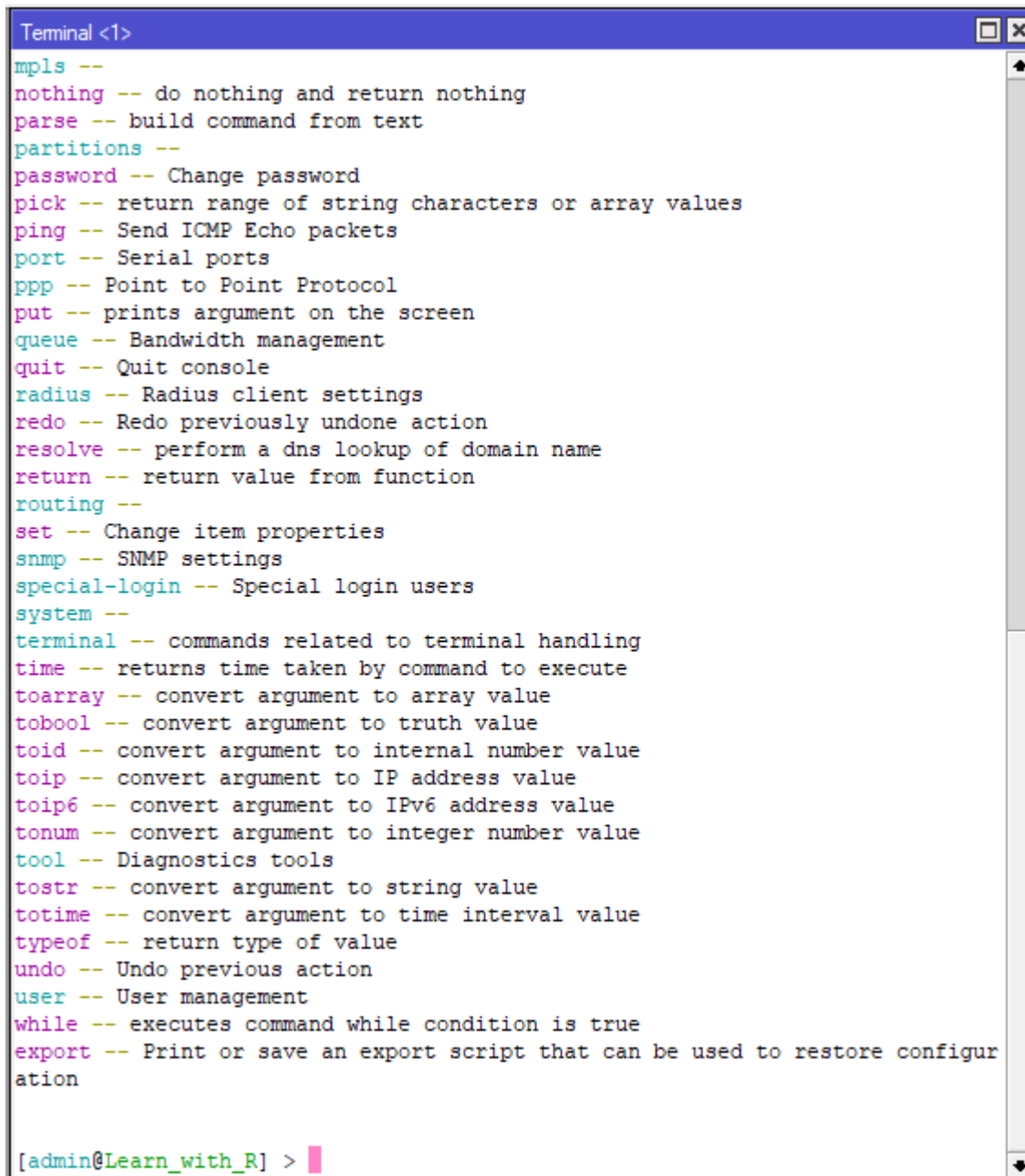


```
Terminal <1>

[admin@Learn_with_R] >
caps-man      interface  mpls         radius        tool         import       undo
certificate   ip         partitions   routing       user         password
console       ipv6       port         snmp          beep         ping
disk          log        ppp          special-login blink        quit
file          metarouter queue        system        export       redo
[admin@Learn_with_R] >
/             do         foreach      nothing       return       toid         totime
:             error        global      parse         set          toip        typeof
environment   execute    if          pick          time         toip6       while
terminal      find       len         put           toarray      tonum
delay         for        local       resolve       tobool       tostr
[admin@Learn_with_R] >
```

- Perintah untuk mengetahui fungsi semua perintah yang ada di mikrotik

Perintah nya : tekan tombol tanda tanya di keyboard



```

Terminal <1>
?
mpls --
nothing -- do nothing and return nothing
parse -- build command from text
partitions --
password -- Change password
pick -- return range of string characters or array values
ping -- Send ICMP Echo packets
port -- Serial ports
ppp -- Point to Point Protocol
put -- prints argument on the screen
queue -- Bandwidth management
quit -- Quit console
radius -- Radius client settings
redo -- Redo previously undone action
resolve -- perform a dns lookup of domain name
return -- return value from function
routing --
set -- Change item properties
snmp -- SNMP settings
special-login -- Special login users
system --
terminal -- commands related to terminal handling
time -- returns time taken by command to execute
toarray -- convert argument to array value
tobool -- convert argument to truth value
toid -- convert argument to internal number value
toip -- convert argument to IP address value
toip6 -- convert argument to IPv6 address value
tonum -- convert argument to integer number value
tool -- Diagnostics tools
tostr -- convert argument to string value
totime -- convert argument to time interval value
typeof -- return type of value
undo -- Undo previous action
user -- User management
while -- executes command while condition is true
export -- Print or save an export script that can be used to restore configur
ation

[admin@Learn_with_R] >
  
```

- Cara menyingkat perintah

```

[admin@Learn_with_R] >
[admin@Learn_with_R] > ip address print
Flags: X - disabled, I - invalid, D - dynamic
#   ADDRESS          NETWORK    INTERFACE
0   10.10.10.5/24     10.10.10.0 ether2-WAN
[admin@Learn_with_R] >
[admin@Learn_with_R] > ip ad pr
Flags: X - disabled, I - invalid, D - dynamic
#   ADDRESS          NETWORK    INTERFACE
0   10.10.10.5/24     10.10.10.0 ether2-WAN
[admin@Learn_with_R] >

```

- Perintah kembali ke directory sebelum nya

```

admin@Learn_with_R] > ip address
admin@Learn_with_R] /ip address> ..
admin@Learn_with_R] /ip> ..
admin@Learn_with_R] >

```

Kesimpulan pembelajaran hari ini :